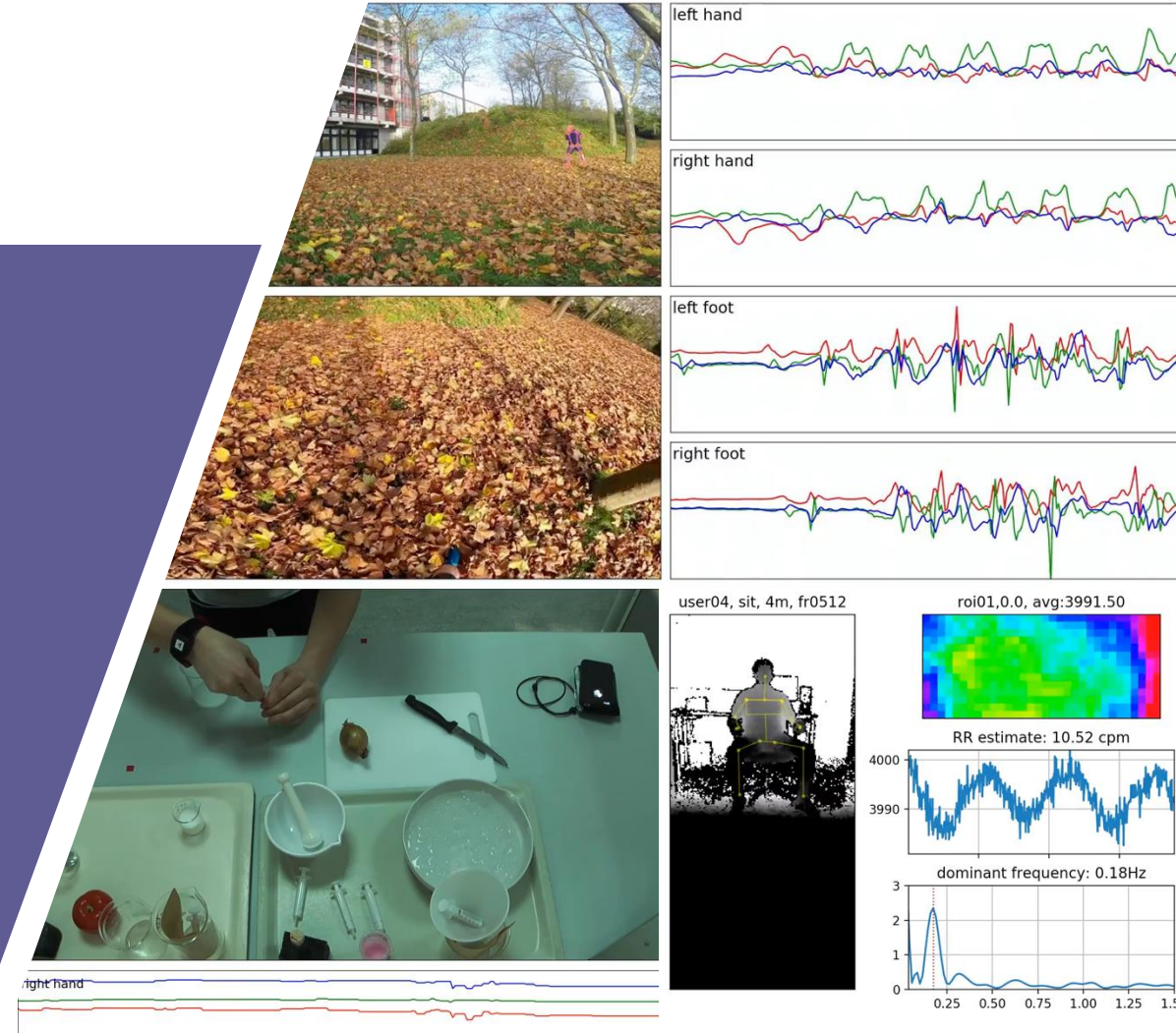


The Data Dilemma in Designing Low-Cost, High- Precision Wearables for Users' Wellness Tracking

Kristof Van Laerhoven

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Exercises Ubiquitous Computing



Upcoming..

1. Wearable Sensing: A Timeline
2. Recording Sensor Data Ubiquitously unleashes Datasets
...but dangers lie in 'blindly' trusting data and annotations
3. Examples of Datasets
4. Visual Inspection of Timeseries, a Recipe
5. Outlook on current research



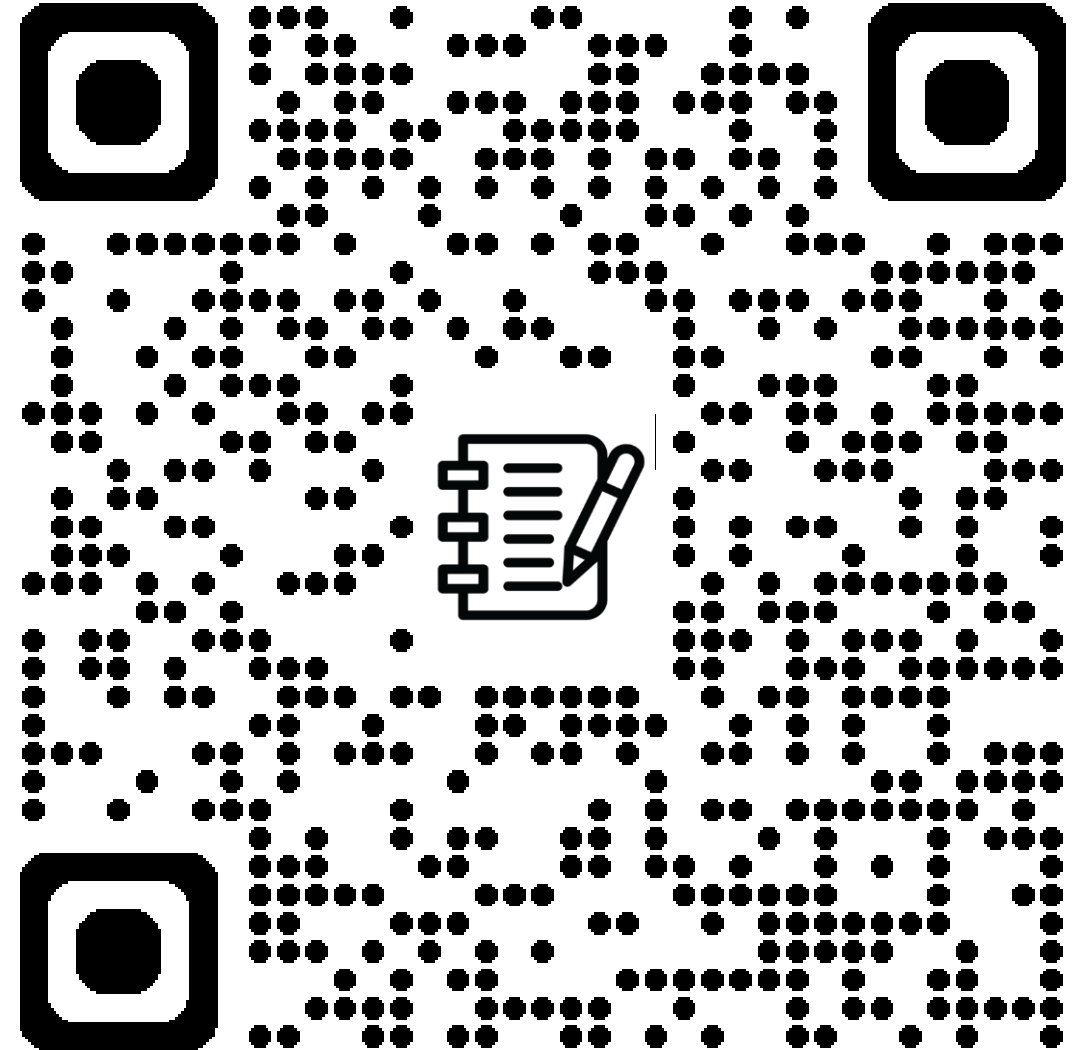
Prepare for some interactivity

<https://shorturl.at/NYTFb>

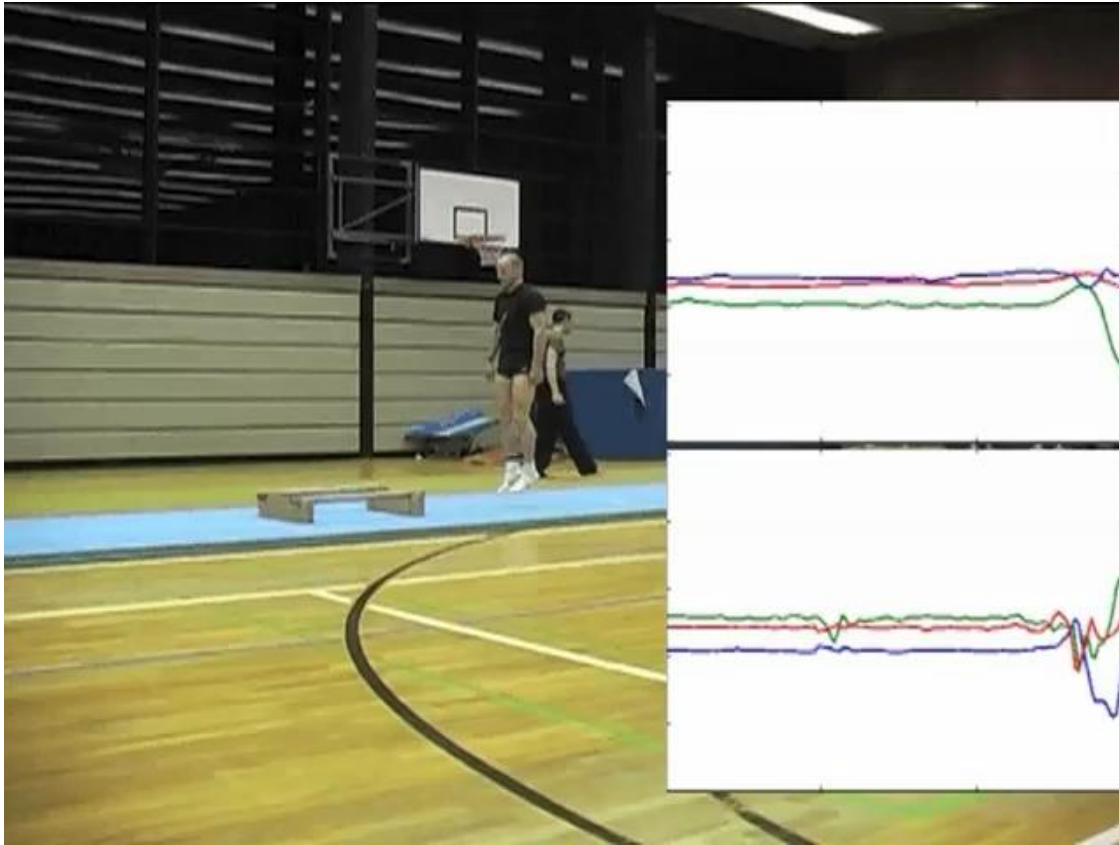
- 1) Does it load?
- 2) Logged into your google account?

👍 That's all for now

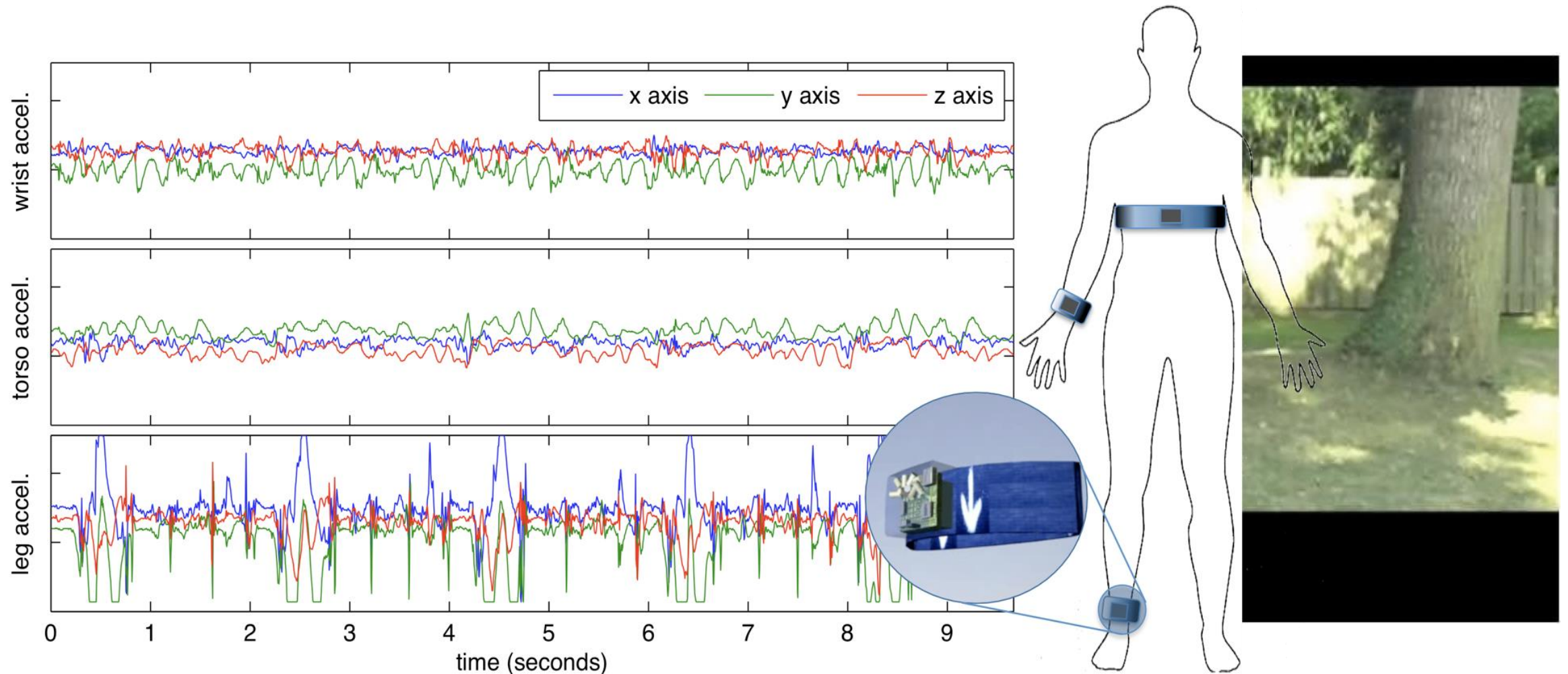
Coding expertise: ★☆☆☆☆



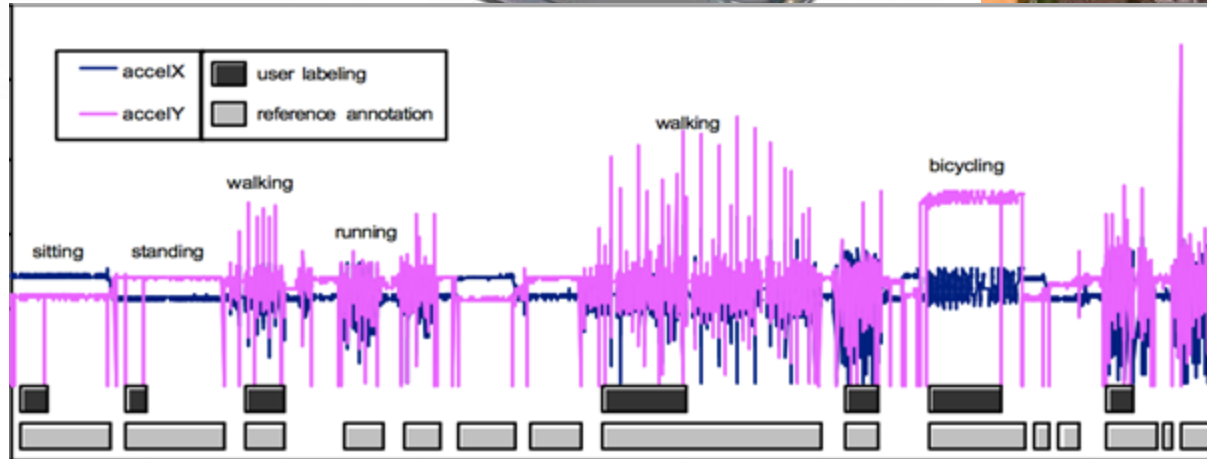
A Timeline: Recording Human Activities



A Timeline: Recording Human Activities

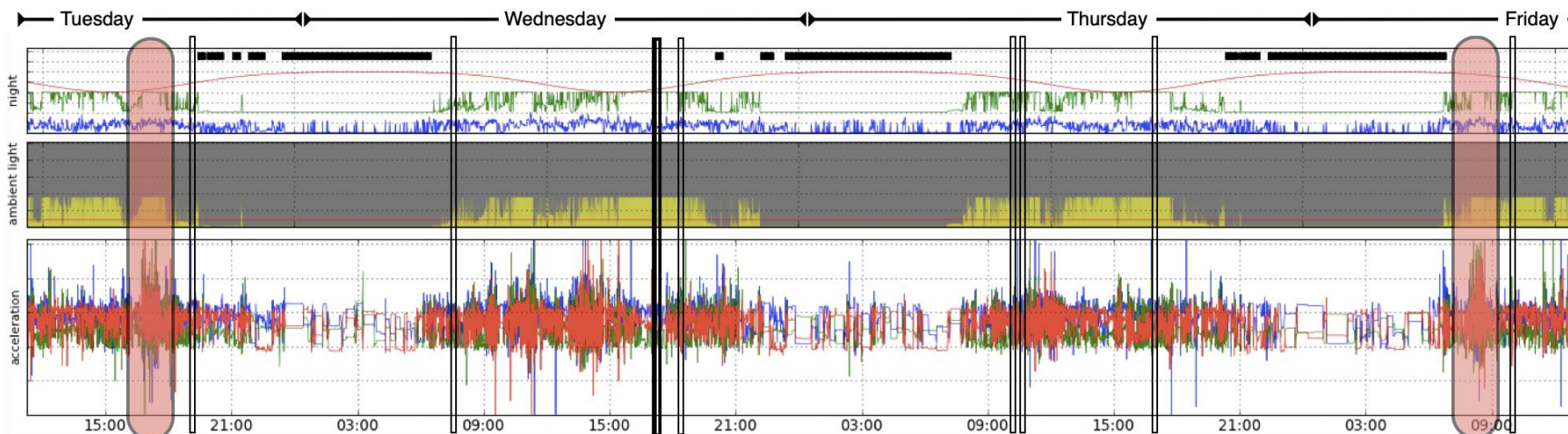
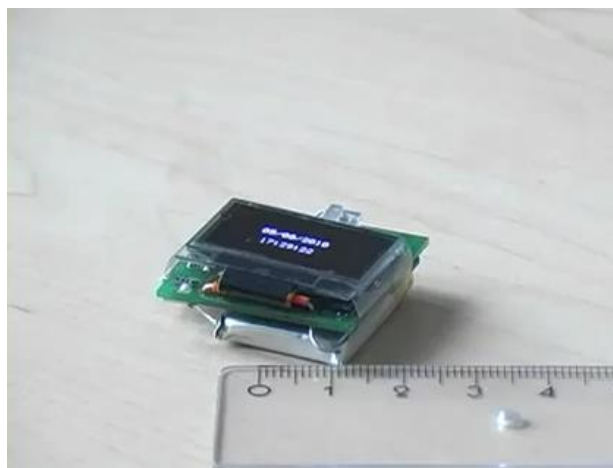


A Timeline: Recording Human Activities: 2000

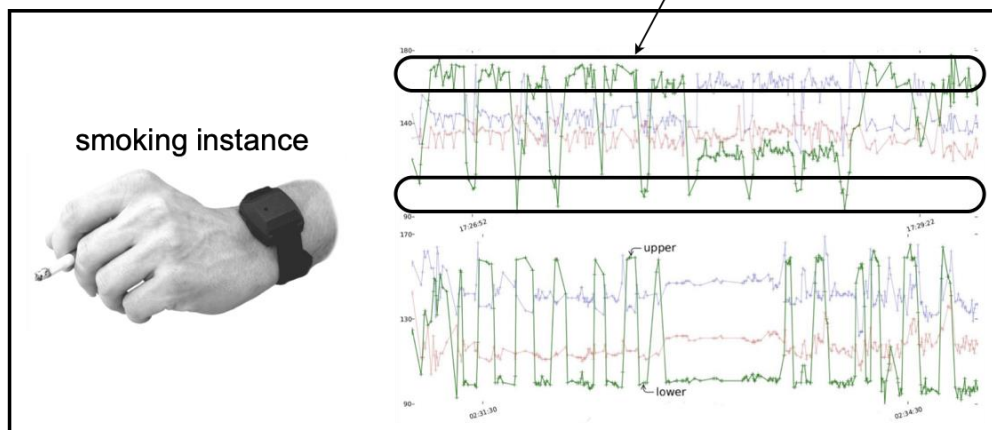


Kristof Van Laerhoven & Ozan Cakmakci,
“What shall we teach our pants?”,
ISWC 2000

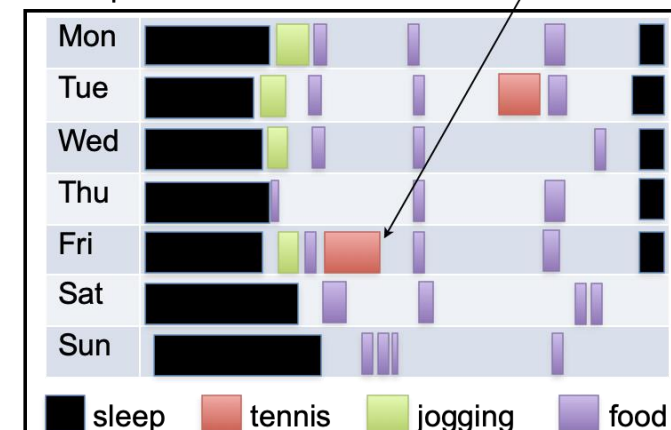
A Timeline: Recording Human Activities: 2012



Gestures



Complex Activities



A Timeline: Recording Human Activities: 2024



<https://kristofvl.github.io/BangleStream>



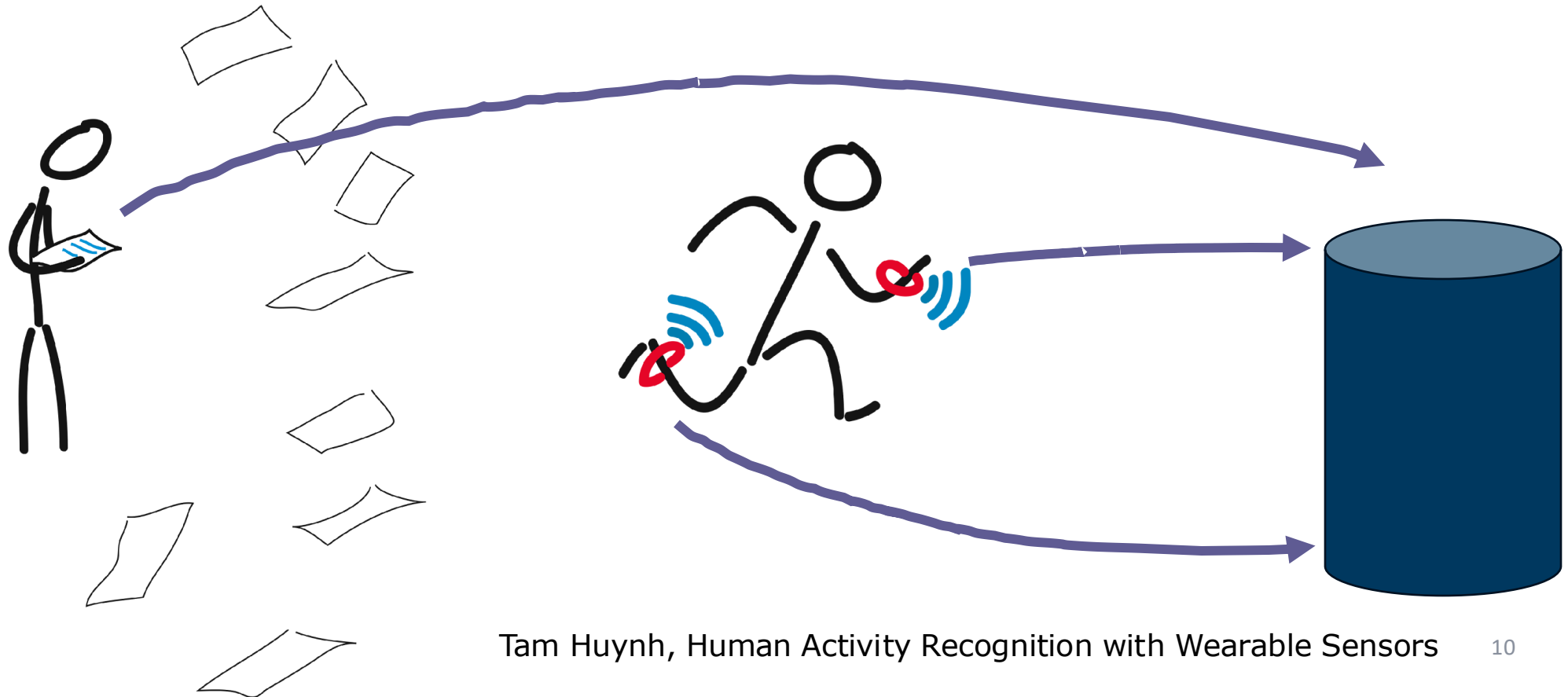
Gordon Williams, Bangle.js: <https://www.youtube.com/watch?v=73aBUkLissM>



A Timeline: Recording Human Activities



User Studies with Wearables



Example 1: Stress and Affect

Study with 15 (3 female) participants, subject to three affective states:

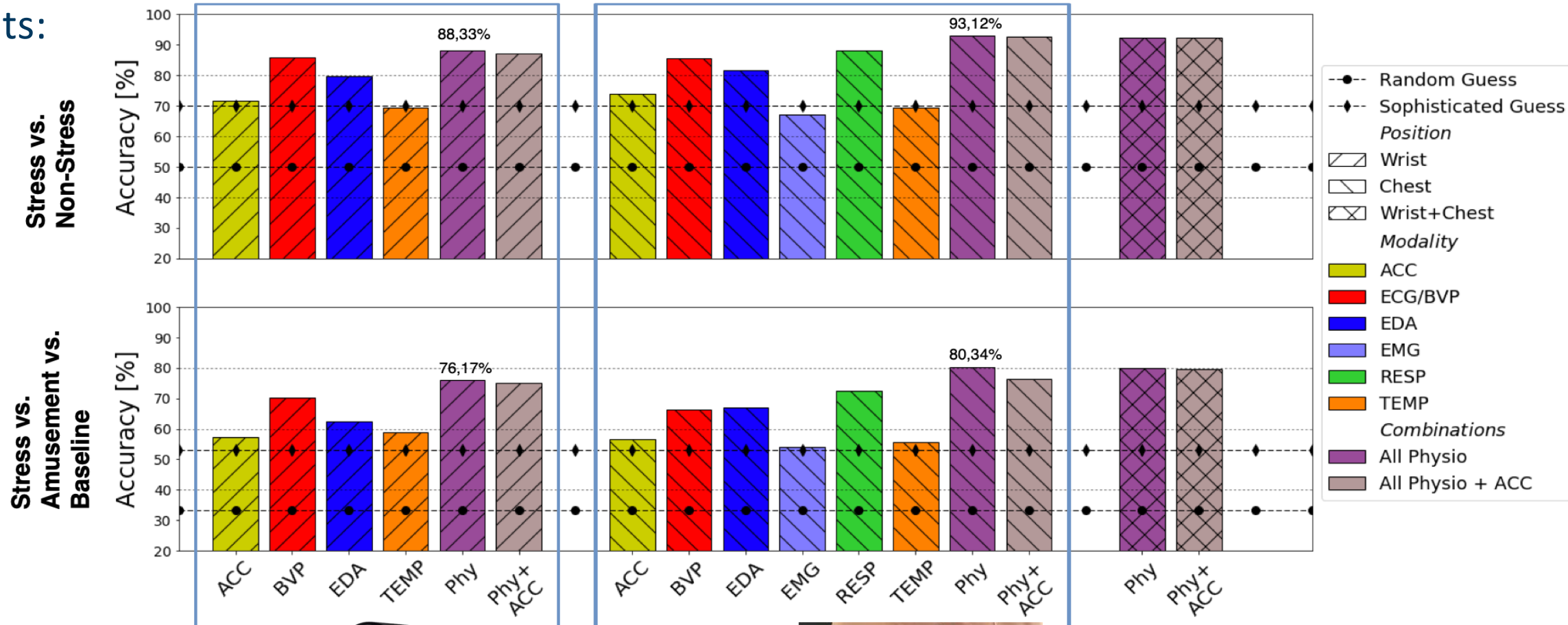


Access our [dataset online](#) or further reading:

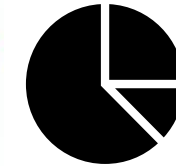
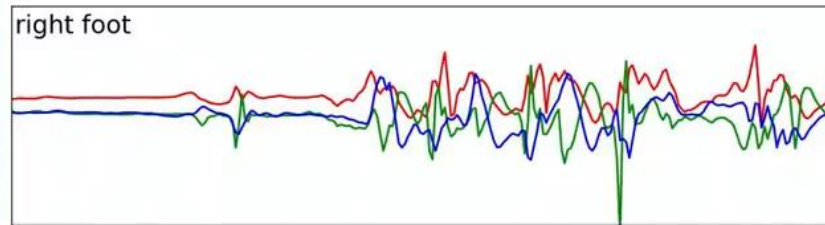
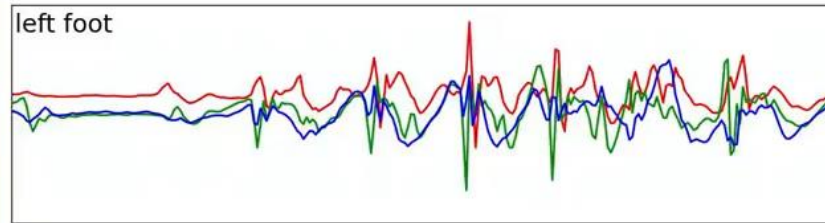
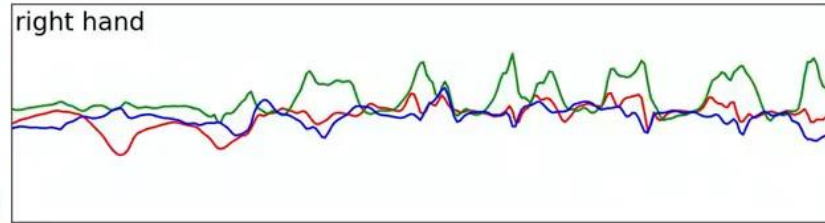
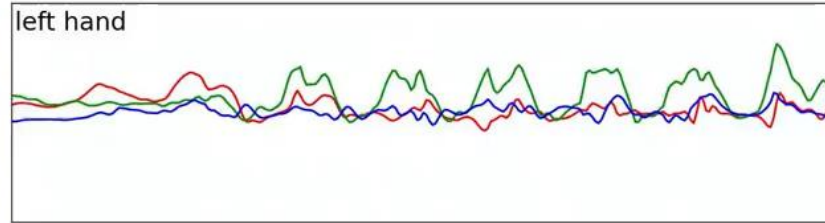
Introducing WESAD, a Multimodal Dataset for Wearable Stress and Affect Detection, Philip Schmidt, Attila Reiss, Robert Duerichen, Claus Marberger and Kristof Van Laerhoven. In ICMI 2018

Example 1: Stress and Affect

Results:



Example 2: WEAR: An Outdoor Sports Dataset



> 19 hours
751 action segments
11 outdoor locations
4 seasons



22 participants
♂ 13 ♀ 9
28 ± 5 years old



min. 2 sessions
18 workout activities
5 running, 5 stretching &
8 strength



WEAR: An Outdoor Sports Dataset for Wearable and Egocentric Activity Recognition, Marius Bock, Hilde Kuehne, Kristof Van Laerhoven, and Michael Moeller. ACM IMWUT, Dec. 2024 <https://mariusbock.github.io/wear/>

Example 2: WEAR: An Outdoor Sports Dataset

Complex Variations

In-Activity Sequences

Little information overlap
across modalities

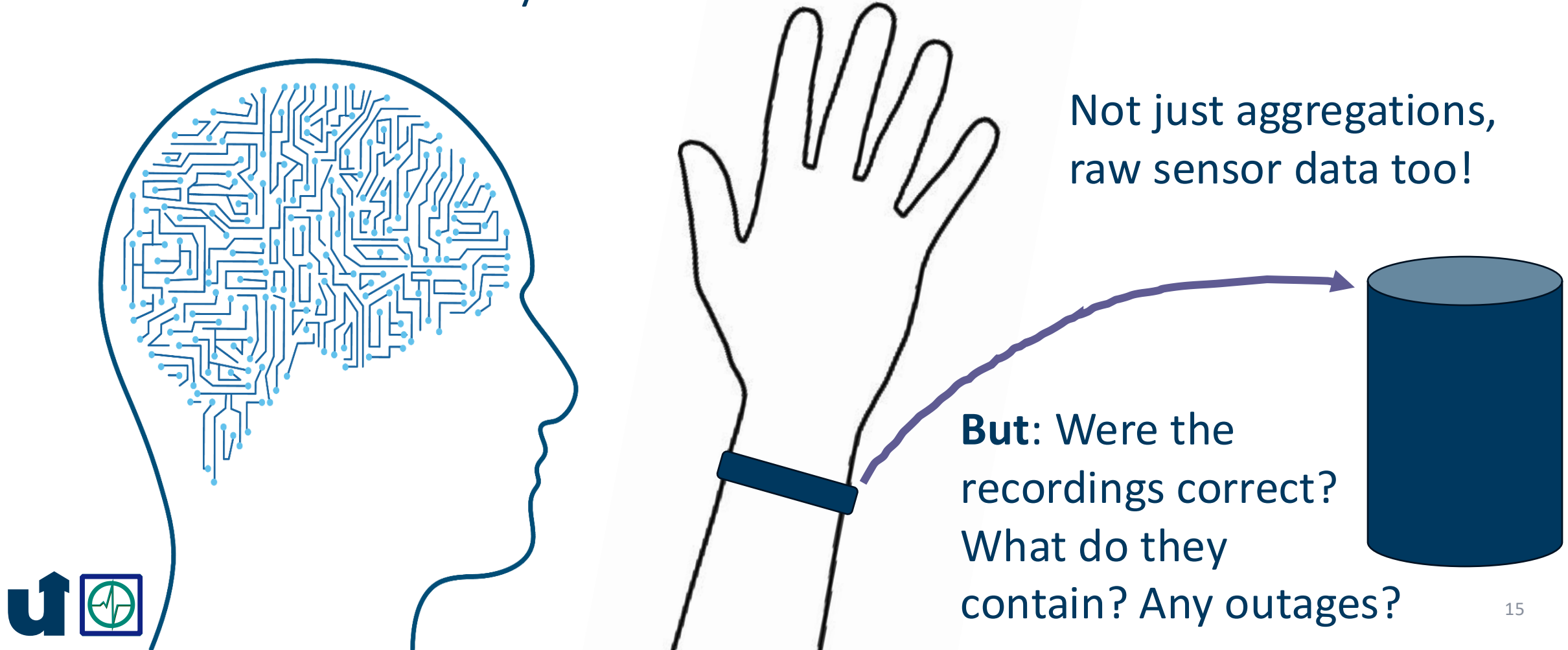


WEAR: An Outdoor Sports Dataset for Wearable and Egocentric Activity Recognition, Marius Bock, Hilde Kuehne, Kristof Van Laerhoven, and Michael Moeller. ACM IMWUT, Dec. 2024 <https://mariusbock.github.io/wear/>

2025 Kaggle
Participation
72 Entrants
33 Participants
25 Teams
630 Submissions

Wearable Data: Mind the Gap

It has become increasingly easier to collect wearable sensor data for your studies



Beyond Data Errors: Label Errors

MNIST

70,000 images of handwritten digits
from 0 to 9

ImageNet

14,197,122 images with 21841
descriptions (synsets)

Many others:

<https://arxiv.org/pdf/1912.05283>



The screenshot shows the ImageNet website interface. At the top, the 'IMAGENET' logo is displayed next to a search bar and a 'SEARCH' button. Below the logo, it states '14,197,122 images, 21841 synsets indexed'. Navigation links for 'Home', 'About', 'Explore', and 'Download' are visible. A user status bar indicates 'Not logged in. Login | Signup'.

The main content area is titled 'Wimp, chicken, crybaby' with the definition 'A person who lacks confidence, is irresolute and wishy-washy'. It shows '290 pictures' and an '81.67% Popularity Percentile'. A 'Wordnet IDs' icon is also present.

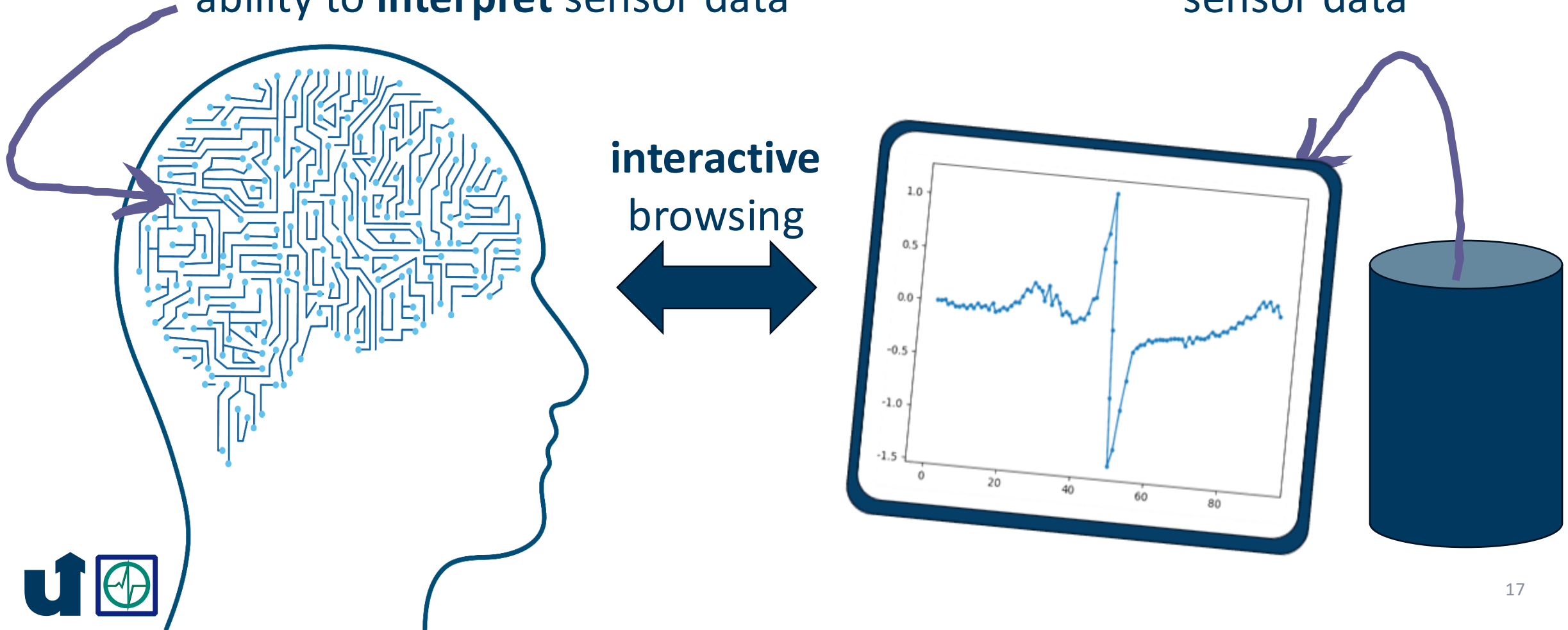
On the left, a list of synsets is shown, with 'wimp, chicken, crybaby (0)' highlighted. The main area displays a grid of image thumbnails under the 'Images of the Synset' tab. A 'Still working...' message is visible above the grid. At the bottom, a pagination bar shows 'Prev', '1', '2', '3', '4', '5', '6', '7', '8', '9', '10', '...', '12', '13', and 'Next'. A disclaimer at the bottom states: 'Images of children synsets are not included. All images shown are thumbnails. Images may be subject to copyright.'

Wearable Data: Mind the Gap

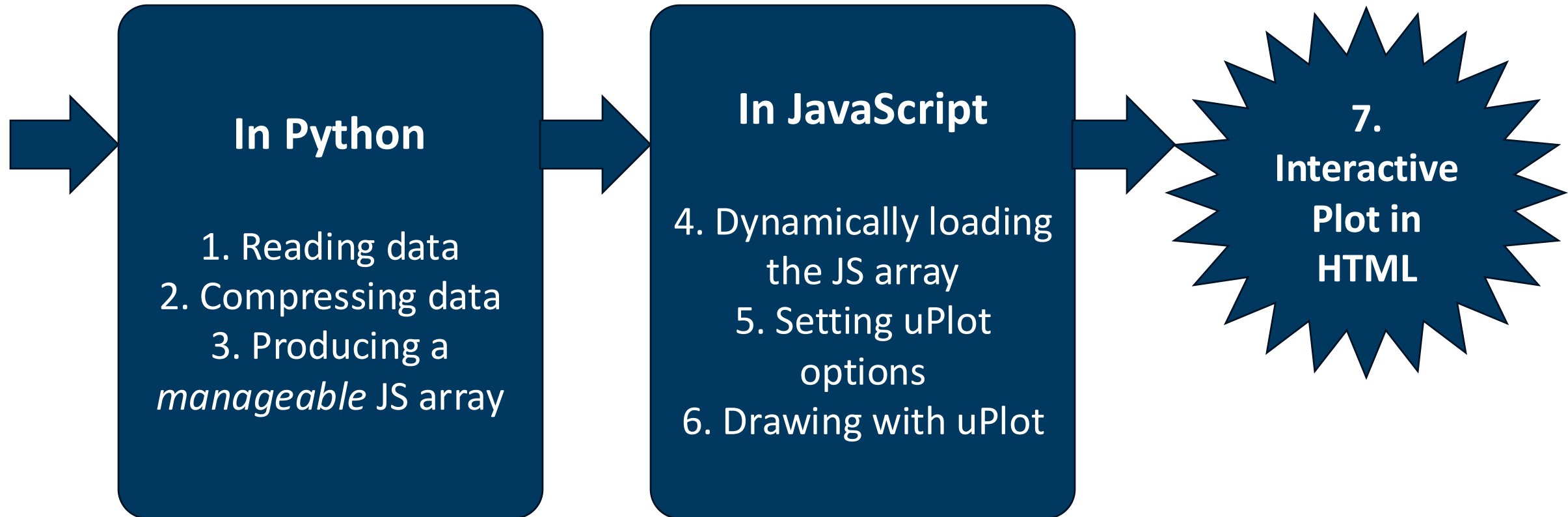
Needed are:

ability to **interpret** sensor data

ability to **inspect**
sensor data



A Basic Recipe in 7 Steps



A Basic Recipe

Let's go back to our earlier notebook:

<https://shorturl.at/NYTFb>



Homework: Let's Go Data Hunting..

WESAD: <https://kristofvl.github.io/wesadviz>

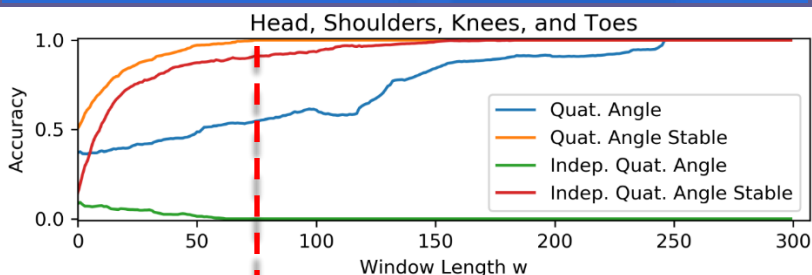
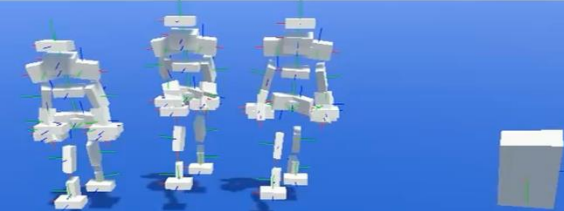
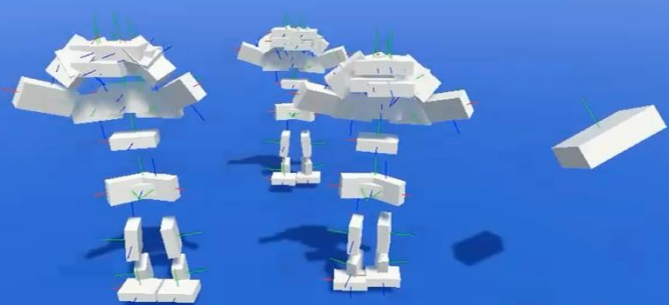
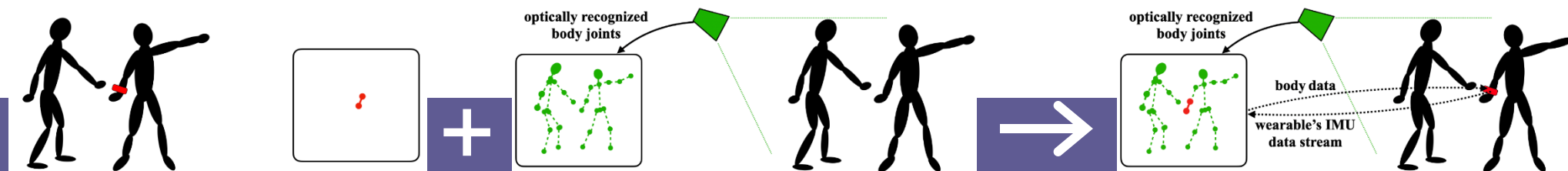
- Which person's chest sensors had a short outage?
- Which persons' heart beats show up in the chest accelerometer data?

WEAR: <https://kristofvl.github.io/wearviz>

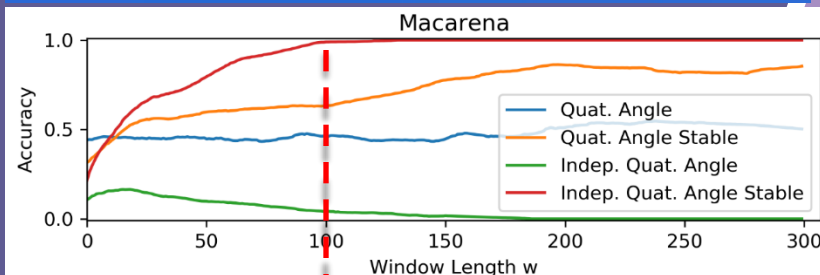
- Which person has missing sensor data and annotations?
- Which person has wrong annotations?



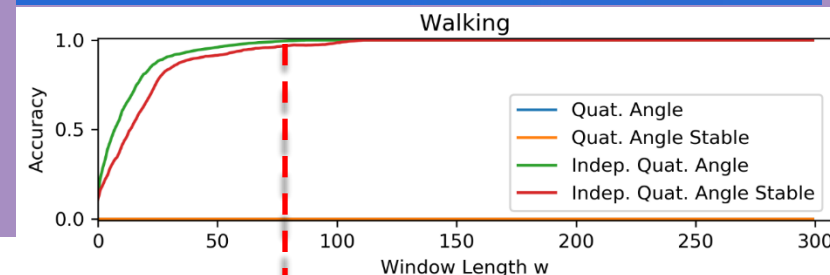
Outlook (1): Using Sensors in the Environment



Perfect match within ~2.5 sec @ 30 Hz



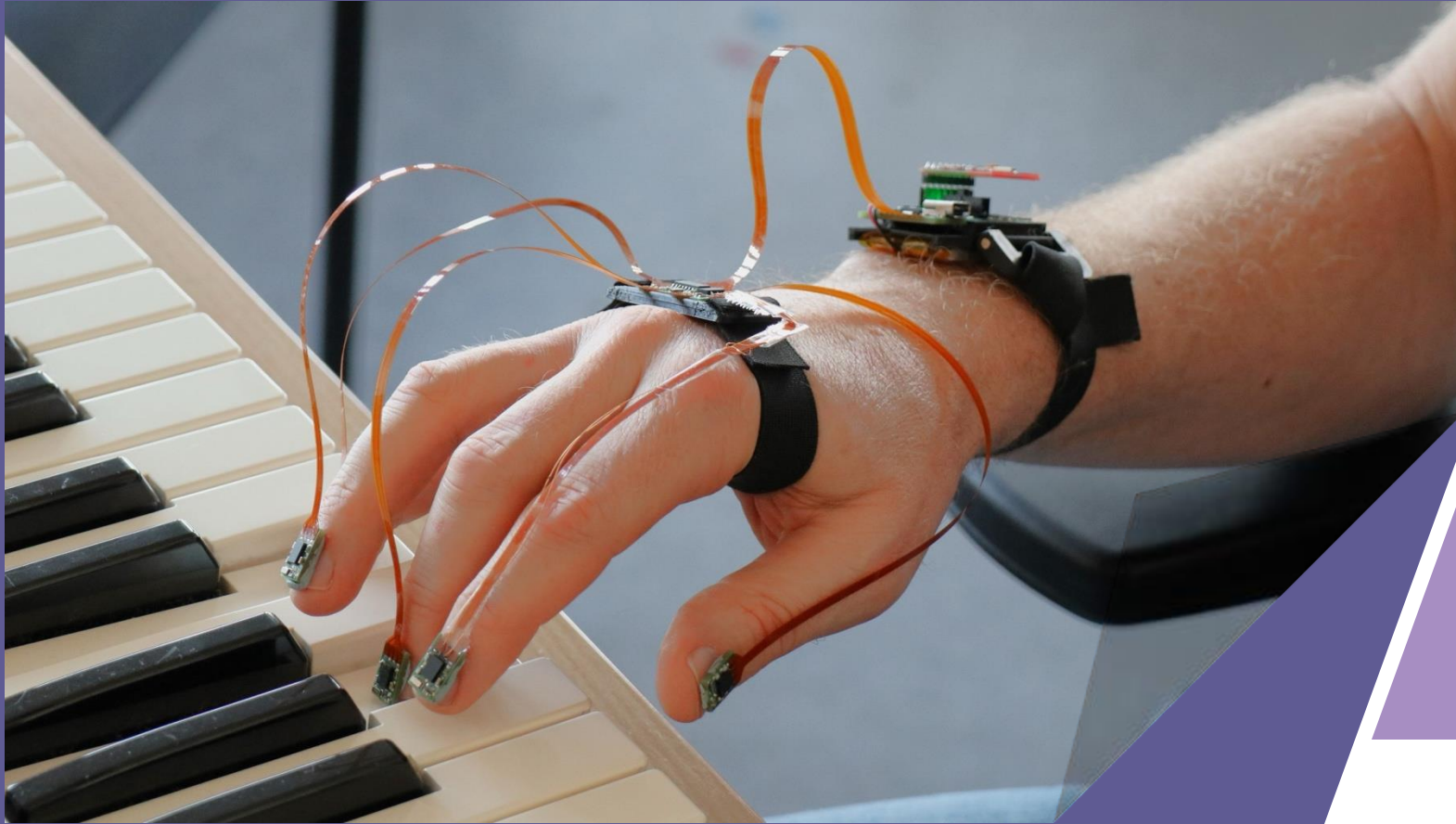
Perfect match within ~3.3 sec @ 30 Hz



Perfect match within ~2.7 sec @ 30 Hz

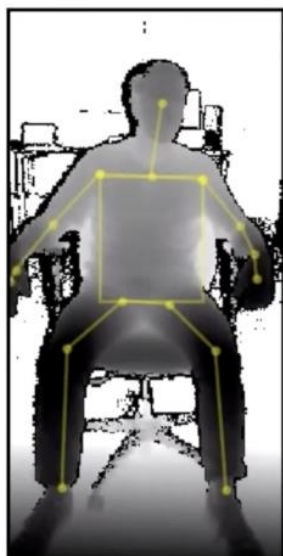


Outlook (2): Miniaturization



Outlook (3): Complementary Modalities

The dataset:
different activities, breathing rates



sitting
(back support)

Towards Breathing as a Sensing Modality in
Depth-Based Activity Recognition, Jochen
Kempfle & Kristof Van Laerhoven, 2020

Thank you for your attention happy to answer questions



<https://ubicomp.eti.uni-siegen.de>



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